

Adnan Harun Dogan

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EDUCATION

-  **Master of Science in Computer Engineering** September 2022 – January 10, 2025
Middle East Technical University (METU) *Ankara, Turkey*
-  **Bachelor of Science in Computer Engineering** September 2, 2018 – July 17, 2022
Middle East Technical University (METU) *Ankara, Turkey*

EXPERIENCE

-  **Research Intern** December 2024 – March 2025
Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich *Zürich, Switzerland*
- Developed EEG-based cybersickness detection using multitaper spectrum estimation in VR
 - Achieved 30% improvement over baselines with novel deep learning architectures
 - Published at IEEE/ICRA 2026
-  **Graduate Research Assistant** February 2022 – Present
Image Processing and Pattern Recognition Laboratory (ImageLab), METU *Ankara, Turkey*
- Designed bucketed ranking loss with Sinkhorn-Knopp algorithm for DETR object detection architecture
 - Improved end-to-end object detection via optimal transport-based loss formulations
-  **Undergraduate Research Intern** June 2021 – September 2021
Cyber-Physical Systems Laboratory, University of California, Irvine *Irvine, California, USA*
- Built multi-modal deep learning system for cardiac and sleep abnormality detection
 - Designed beat-by-beat arrhythmia classification integrating time-series and spectral features
 - Published at Computing in Cardiology (CinC) 2021

PUBLICATIONS

- Dogan, A. H.**, Demirel, B. U., Holz, C. (2026). Frequency-Weighted Neural Kalman Filters. *ICRA 2026*. [GitHub]
- Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus M., Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using eeg-based multitaper spectrum estimation. *IEEE/TVCG 2025*. [DOI — GitHub]
- Yavuz, F., Cam B.C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., Akbas, E. (2024). Bucketed Ranking Loss for Efficient Ranking-based Training of Object Detectors. *ECCV 2024*. [ECVA — DOI — GitHub]
- Amin M. A., **Dogan, A. H.**, Kuru E. S., Sever Y., Angin P. (2024). Misuse Detection and Response for Orchestrated Microservices Based Software. *AINA 2024*. [KAGGLE — PAPER]
- Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, S. T., Tunel, B. M. (2024). Enhancing Underground Built Heritage Analysis with Text Mining: A Case Study on Cappadocia. *Book Chapter 2024*. [PAPER]
- Sever Y., **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. *ITU Journal 2023*. [DOI — Paper — GitHub]
- Sever, Y., Ekinici, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., Angin, P. (September 2022). An Empirical Analysis of IDS Approaches in Container Security. *IEEE/SRMC 2022*. [DOI]
- Demirel, B. U., **Dogan, A. H.**, Al-Faruque, M. A. (2021). Two Might Do: A Beat-by-Beat Classification of Cardiac Abnormalities using Deep Learning and Domain-Specific Features. *CinC 2021*. [GitHub — PAPER — DOI]
- Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., Karagoz, P. (2021). Explainability in Irony Detection. *DaWaK 2021*. [PAPER]
- Dogan, A. H.**, Dogan, A. (June 2021). An Assembled Deep Learning Approach for Flow Field Prediction. [PAPER]

HONORS & AWARDS

-  **TUBITAK 2247 C STAR Scholarship** 2022-2023
The Scientific and Technological Research Institution of Turkey (TÜBİTAK)
-  **Best Paper Award** September 2022
IEEE/SRMC'22
-  **TUBITAK 2247 C STAR Scholarship** 2021-2022
The Scientific and Technological Research Institution of Turkey (TÜBİTAK)
-  **TUBITAK 2247 C STAR Scholarship** 2020-2021
The Scientific and Technological Research Institution of Turkey (TÜBİTAK)

TECHNICAL SKILLS

Skills: Python/PyTorch, Variational and Generative Modeling, Computer Vision, Statistical Machine Learning, Combinatorial Optimization, Reinforcement Learning